

STAT293 - Examples

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Step 1: Source

```
rm(list = ls())
graphics.off()
gc()

source("00_requirements.R")
source("01_DataSimulation.R")
source("12_Method1_aux.R")
source("21_Method2.R")
source("22_Method2_aux.R")
source("3_Evaluation.R")
```

Step 2: Simulation Paramaters

```
set.seed(234)

# Dimensions
K <- 10      # Number of individuals
T_len <- 30   # Total number of time points
d <- 10      # Number of variables (Dimension)

# Model Structure
p <- 1       # VAR lag order
sparsity <- 0.05 # Sparsity level (proportion of non-zero coefficients)
heterogeneity <- "medium" # Level of heterogeneity ("low", "medium", "high")

# Data Generation & Noise
sigma_ep <- 0.1 # Standard deviation of noise
phi_max <- 0.9  # Upper bound for coefficient generation
phi_min <- 0.1  # Lower bound for coefficient generation

# Evaluation Settings
n_holdout <- 5 # Number of time points withheld for forecasting (holdout)

# Code below (do not modify)
# Heterogeneity
frac_common <- switch(heterogeneity,
                      "low" = 2/3,
```

```

        "medium" = 1/2,
        "high"   = 1/3)

# total number of nonzero elements in each  $d \times d$  transition matrix
n_nonzero_total <- max(1, round(sparsity * d * d))

# number of nonzero elements for common effects
n_common <- max(1, round(frac_common * n_nonzero_total))

# number of nonzero elements for unique (individual-specific) effects
n_unique <- n_nonzero_total - n_common

```

Step 3: Generate $\text{Var}(p)$ ILD for each individual

```

# Generate Data for All Subjects
# data_list[[k]] contains the observed data matrix for subject k
# individual_B is 4D array [d, d, p, K]

individual_B <- gen_individual_B(K, d, p, n_common, n_unique)

data_list <- lapply(1:K, function(k) {

  # Get coefficients for subject k
  # Dimensions: [d, d, p]
  subject_phi <- array(individual_B[, , k], dim = c(d, d, p))

  # Generate time series data
  # Returns a matrix of size [T_len, d]
  gen_VAR_p(subject_phi, T_len, p, d, sigma_ep)

})

```

Step 4: Split Training & Test Datasets

```

# Define the split index (The end of the training period)
split_idx <- T_len - n_holdout

# Create Training Data
train_data <- lapply(data_list, function(X) { X[1:split_idx, ] })

# Create Test Data (With Lookback)
# Include the last 'p' rows of the training set.
# The "initial conditions (lags)" for the first prediction in the test set.
test_data <- lapply(data_list, function(X) { X[(split_idx + 1 - p):T_len, ] })

# --- Note for Evaluation ---
# 'test_data' now has (n_holdout + p) rows. The first 'p' rows are history.
# Evaluate metrics only on the last 'n_holdout' rows (the actual test set).

```

Step 5: (a) Individual LASSO for VAR(p)

Fit Lasso VAR(1) and VAR(p) for each subject

```
# VAR(p)
if (p != 1){
  est_lasso_B_p <- lapply(train_data, function(data) {
    fit_lasso_var_p(data, p)
  })

  est_lasso_B_p <- array(unlist(est_lasso_B_p), dim = c(d, d, p, K))
}

# VAR(1)
est_lasso_B <- lapply(train_data, function(data) {
  fit_lasso_var_p(data, p = 1)
})

est_lasso_B <- array(unlist(est_lasso_B), dim = c(d, d, K))
```

Step 6: (b) Multi-VAR

Standard LASSO

```
#lambda_standard <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_lasso_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_standard,
  #lambda2 = lambda_standard,
  standardize = TRUE,
  canonical = FALSE,
  lassotype = "standard", # standard multi-VAR (not adaptive)
  cv = "blocked",
  nfolds = 10
)

multivar_lasso_res <- cv.multivar(multivar_lasso_model)
```

```
##      |
est_multivar_lasso_B <- multivar_B(multivar_lasso_res$mats$total, d, K)
```

Adaptive

```
#lambda_adaptive <- 10^seq(log10(0.01), log10(5), length.out = 30)

multivar_adapt_model <- constructModel(
  data = train_data,
  lag = 1,
  #lambda1 = lambda_adaptive,
  #lambda2 = lambda_adaptive,
```

```

standardize = TRUE,
#weightest = "ridge",
canonical = FALSE,
lassotype = "adaptive",
cv = "blocked",
nfolds = 10
)

multivar_adapt_res <- cv.multivar(multivar_adapt_model)

## |

est_multivar_adapt_B <- multivar_B(multivar_adapt_res$mats$total, d, K)

```

Step 7: Evaluation

Estimation Errors

Table 1: Comparison of Estimation Error

Method	Error_Metric
LASSO (Lag 1)	11.386
Multi-VAR (Standard)	6.324
Multi-VAR (Adaptive)	11.475

Sensitivity and Specificity

Table 2: Comparison of Mean Sensitivity and Specificity

Method	Sensitivity	Specificity
LASSO (Lag 1)	0.70	0.843
Multi-VAR (Standard)	0.76	0.840
Multi-VAR (Adaptive)	0.20	0.997

Sensitivity and Specificity (per subject)

Table 3: Model Performance – Mean (SD) of Sensitivity and Specificity across Subjects

Method	Sensitivity Mean (SD)	Specificity Mean (SD)
LASSO (Lag 1)	0.700 (0.216)	0.843 (0.087)
Multi-VAR (Standard)	0.760 (0.207)	0.840 (0.039)
Multi-VAR (Adaptive)	0.200 (0.267)	0.997 (0.005)

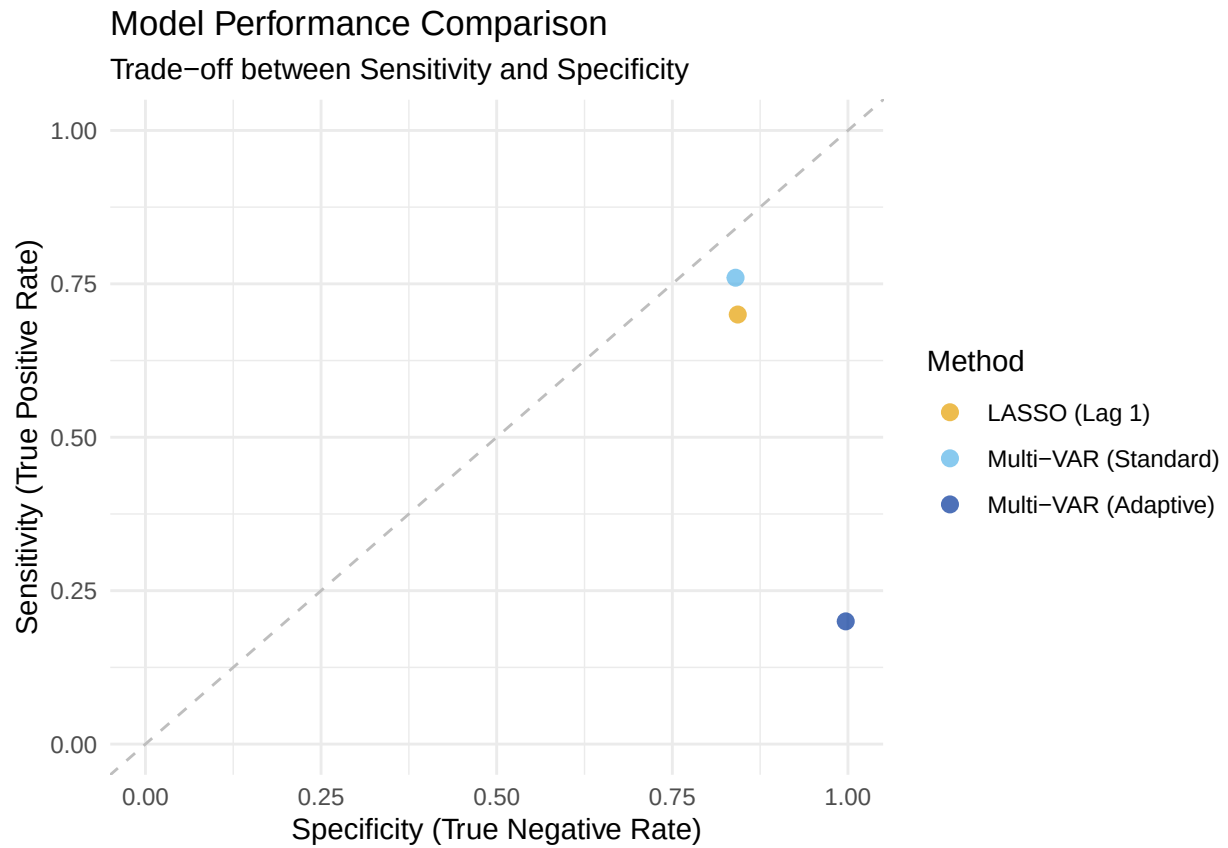
RMSFE (Root Mean Square Forecast Error)

Table 4: Model Performance – Mean (SD) of RMSFE across Subjects

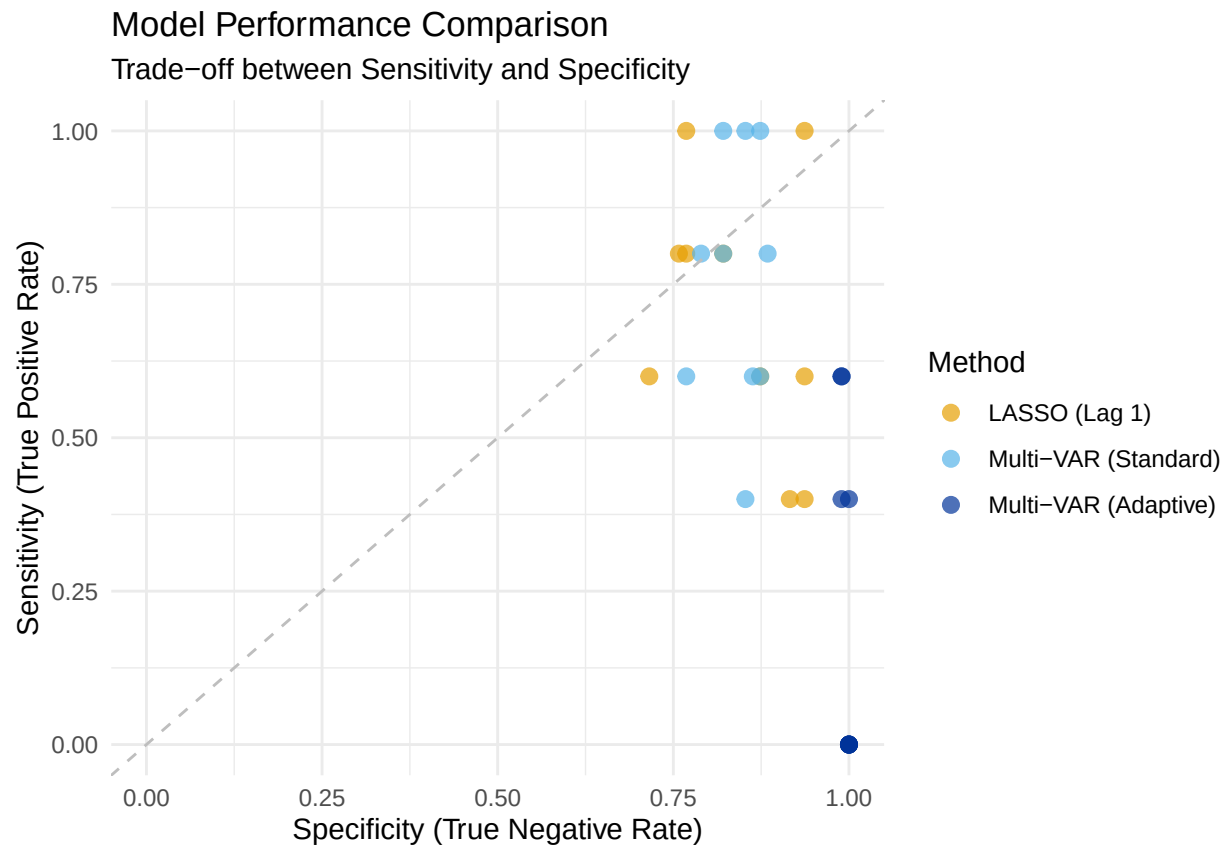
Method	RMSFE Mean (SD)
LASSO (Lag 1)	0.106 (0.013)
Multi-VAR (Standard)	0.103 (0.012)
Multi-VAR (Adaptive)	0.105 (0.013)

Step 8: Visualization

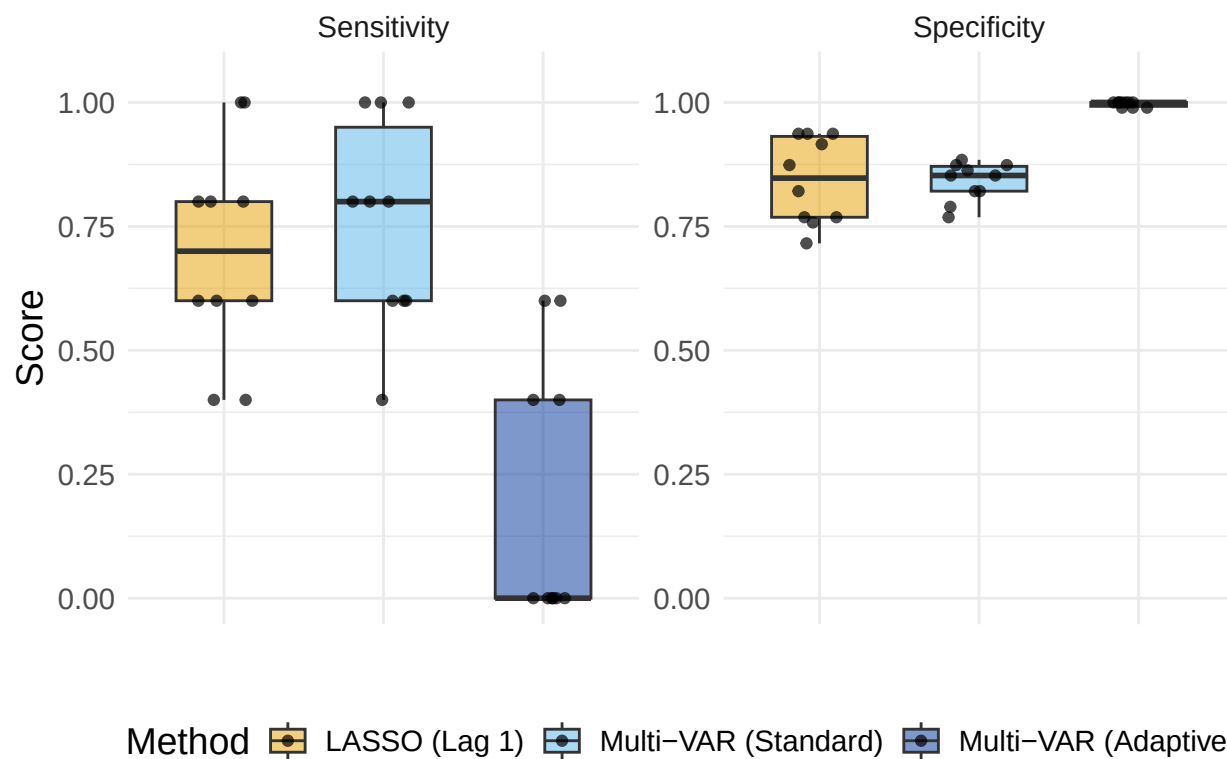
Sensitivity and Specificity (mean)



Sensitivity and Specificity (per subject)



Model Performance Distribution (Per Subject)



RMSFE (per subject)

